

Stephen Qiu

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EDUCATION

Masters in Robotics

University of Pennsylvania

Aug 2024 - May 2026 (expected)

- GPA: 4.0/4.0

Bachelor of Science in Mechanical Engineering

University of Tennessee, Knoxville

Aug 2020 - May 2024

- GPA: 3.98/4.0, Summa Cum Laude
- Minor in Computer Science

RESEARCH EXPERIENCE

Undergraduate Researcher

REACH Robotics Lab

Jan 2023 - May 2024

University of Tennessee, Knoxville

- Collaborated with PhD researchers to design a mock surgical system setup where I focused on implementing code for reading inputs from a haptic input device as well as setting up the communication interface between each of the setup's components using ROS.
- Designed a Python interface into MATLAB to help with the geometry creation of concentric push-pull robots as part of a larger finite element analysis pipeline. Increased overall code speed by over 60 percent.
- Worked on the optimization of concentric push-pull robot designs using FEA and MATLAB. Led to discovery of new robot design parameters with better stiffness properties.

Undergraduate Researcher

National Institute for Computational Sciences

Jan 2021 - May 2024

University of Tennessee, Knoxville

- Researched and contributed to MagmaDNN, an open source C++ neural network framework where I implemented and optimized machine learning algorithms and models such as Resnet and UNet into the framework using NVIDIA's CUDA and CUDNN packages.
- Trained autonomous vehicles using NVIDIA's Jetson platform to perform tasks such as line following and object avoidance using computer vision.
- Organized the LAPENNA workshop teaching around 20 participants about emerging topics in machine learning, computer vision and artificial intelligence.

NSF REU Participant

RECSEM

June 2021 - Aug 2021

University of Tennessee, Knoxville

- Developed an algorithm to do tensor matrix calculations in C++ for a neural network with speeds comparable to TensorFlow.
- Created an interface to exchange numerical data between two different frameworks for better compatibility between the two frameworks.

REFEREED CONFERENCE PUBLICATIONS

1. K.T. Dang, **S. Qiu**, C. Hatch, P. Connor, T. Qin, R. Alterovirz, R.J. Webster III, and C. Rucker, "Design of Transmission Tubes for Surgical Concentric Push-Pull Robots" *IEEE ISMR 2024, Atlanta, Georgia, USA*
2. J. Halloy, **S. Qiu**, S. Tomov, and K. Wong, "PyMAGMA: A Python Interface for MAGMA." *ACM PEARC 2024, Providence, Rhode Island, USA*

SKILLS

Hardware Design

SolidWorks, Fusion 360, Autodesk Inventor

Programming

Matlab, C++, C, Python, Arduino

Other

Linux Systems, ROS, Github, 3D Printing, Microsoft Office